

Lab #3 - Linewidth Measurements

ECE 482/582, Fall 2003

Goals:

1. To observe the spectral line-shapes of several different light sources.
2. To learn to use the Ocean Optics and Chromex spectrometers and associated lab instruments such as the lock-in amplifier, chopper, chart recorder, etc.
3. To gain an intuitive feel for what the line-shape function represents.
4. To observe the emission lines from a HeNe discharge and try to identify both spontaneous and lasing lines and measure the linewidths.
5. To understand the differences between the mechanisms of homogeneous and inhomogeneous broadening.

Reading:

Kuhn, *Laser Engineering*, pp. 34-43

Other Resources:

Verdeyen, *Laser Electronics*, 3rd Ed., 1995, Prentice-Hall, Chapter 7

Moore, John, *Building Scientific Apparatus*, Section 4.8

www.jobinyvon.com/jy/oos/oos1.htm#top - Optics of Spectroscopy Tutorial

Preparatory Questions:

1. Briefly describe (using words and diagrams) the different types of line broadening discussed in the reading; homogeneous, inhomogeneous, Voigt mixed.
2. A monochromator is frequently used to measure the spectral response of a light source. Do some library or web research and describe how a simple grating monochromator works - particularly resolution and dispersion and how gratings and slits affect these.
3. Consider the problem of measuring the linewidth of the HeNe laser using a grating monochromator. What limits the resolution of the measurement?
4. How many lines do you think might come out of the spontaneous side-emission from a HeNe laser tube? What do you think they might be due to?

Laboratory Experiment:

The linewidth of a source, usually a laser, refers to the full width at half maximum (FWHM), of the principle gain profile in the wavelength spectrum. Generally a Lorentzian or a Gaussian distribution can represent the shape of this profile. In theory, if we plot the spectral distribution of a sufficiently wide enough band, we can readily determine the linewidth of the source centered about some wavelength. In practice, the distribution may not necessarily reflect the actual linewidth of that source. In particular, the measuring apparatus may limit the resolution of very narrow sources.

In this lab we will be using the CCD array Ocean Optics spectrometer and the Chromex monochromator with lock-in amplifier. The main challenges in this lab will be optical alignment, coupling of the source into the monochromator, and interpreting your resulting spectra in terms of the monochromator resolution and the predicted linewidths.

Procedure:

Using the existing setup and following the procedures shown to you during the lab demonstration complete the following:

1. Using the CCD array spectrometer (Ocean Optics), measure and sketch in your lab notebook the spectra of the following:
 - a. The overhead fluorescent lights in the lab.
 - b. The LEDs in the demo box (red, green, blue-green, and white).
 - c. The spontaneous side-emission from the HeNe laser tube.
 - d. Reflectance spectra of several different colors of paper.
 - e. The Hg/Ar calibration source for the Ocean Optics spectrometer.
 - f. The He-Ne laser output (look at its scattering from white paper, don't shine the laser right into the fiber!).
2. Use the Chromex 0.5m monochromator with the chopper and lock-in amplifier to get a higher resolution measurement of the HeNe laser spectrum. Vary the slit width, scanning speed, etc., to obtain the best results. Try slits of 100 microns, 30 microns, and 10 microns.

Questions to Answer:

- Sketch a simple diagram of the experimental setups and briefly explain how accurate measurements are achieved for both systems.
- Explain carefully the impact of the slit width on the measurements.
- Draw the spectral distributions for the given sources and indicate the center wavelengths and the associated linewidths for each. Comment on any irregularities in the line-shape.
- Using the value of 1.5GHz for the Doppler linewidth of the HeNe laser output, calculate the corresponding wavelength spread. How does this value compare to the experimental result you measured for the HeNe laser?
- What is the measured minimum resolution of the Chromex with the 10 micron slit width? What can you conclude about the HeNe laser linewidth you measured?
- How does the HeNe laser linewidth compare to the other sources you measured?
- How do the sources you measured compare in line-shape?

Conclusion:

You should now have an understanding of the various line broadening mechanisms that occur in laser sources and how these affect the linewidth and line-shape. You have measured the spectral linewidth of several different sources and should have a feel for the measurement techniques and precision required to do so.